

Juan Spinelli

Software Engineer | Backend, Cloud & Data Engineering

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TECHNICAL PROFILE

- Backend Engineer and aspiring Data/Cloud Engineer building production-grade serverless pipelines, REST APIs, and AI-powered applications across AWS and GCP.
- Multi-cloud practitioner: AWS (Lambda, EventBridge, S3, SES, API Gateway, CloudWatch, DynamoDB, SNS) and GCP (Cloud Functions, Cloud Scheduler), with Azure Cosmos DB for cross-cloud deduplication.
- Experienced integrating LLM APIs (Groq Llama 3.3 70B, Claude) into end-to-end products — from AI-enriched job intelligence pipelines to dual-persona mobile fitness coaching apps.
- Follows a consistent pipeline architecture: Docker → serverless compute → event-driven scheduling → S3 data lake → CI/CD via GitHub Actions → Vercel frontend; all secrets via AWS SSM Parameter Store.
- Self-motivated team contributor comfortable across the stack — from Node.js/Express backends and Prisma ORMs to React dashboards and mobile apps with Expo.

SKILLS

Languages:	Python, JavaScript, TypeScript, Java, SQL, Swift, C++, HTML, CSS
Cloud & DevOps:	AWS (Lambda, EventBridge, S3, SES, API Gateway, CloudWatch, SNS, SSM, DynamoDB), GCP (Cloud Functions, Cloud Scheduler), Azure Cosmos DB, Docker, GitHub Actions (CI/CD), Render, Vercel
Data Engineering:	ETL pipeline design, REST API ingestion, Pydantic validation, S3 data lake patterns, cross-run deduplication, structured CloudWatch logging, EventBridge scheduling, telemetry & alerting
Databases:	PostgreSQL (Supabase), MongoDB, MySQL, Firebase, DynamoDB, Azure Cosmos DB
Frameworks & Libs:	Node.js, Express.js, Prisma, FastAPI, React Native (Expo), React, Vite, Recharts, Pygame, Unity, ArcGIS
AI / LLM:	Groq API (Llama 3.3 70B), Anthropic Claude API, LLM prompt engineering, NLP, vector embeddings
Mobile & XR:	Android SDK, SwiftUI, visionOS SDK, RealityKit, ARKit, Oculus Meta Quest SDK
Tools:	Postman, Swagger, VS Code, Android Studio, Jupyter, Git, Prisma ORM, AWS SSM
Certifications:	Mobile Applications Development (Android & iOS); Swift Programming – Miami Dade College

PROFESSIONAL EXPERIENCE

TriMerge Consulting Group, P.A.

Mar. 2026 – May 2026

Backend Developer Intern

Miami, FL

- **[Node.js, Express, JWT, RBAC, MongoDB]** Developed and maintained RESTful API endpoints across a 9-module backend system supporting multi-user authentication and role-based access control.
- **[Postman, Swagger]** Tested and documented all API endpoints prior to deployment; collaborated with frontend developers to ensure seamless client-server data integration.
- **[Node.js, Python, Firebase, AI/NLP]** Built the TriMerge AI Proposal Builder — an AI-powered tool automating consulting proposal creation using NLP — and contributed to the JWT Auth System with JWT-based secure authentication.
- **[Lovable, PowerPoint, GitHub Actions]** Produced FEMA/hazard mitigation RFQ proposal materials: five-page interactive dashboard, workflow diagram PDF, and PowerPoint pitch for VP-level non-technical stakeholders.
- **[GitHub, Git]** Participated in code reviews and managed version control following branching and pull-request workflows.

City of Coral Gables

Sept. 2023 – Nov. 2023

GIS & IoT Intern

Coral Gables, FL

- **[Unity, ArcGIS, Meta Quest]** Built a virtual reality model of Coral Gables integrating GIS data to produce spatially accurate immersive city visualizations navigable via Meta Quest devices.
 - Worked cross-functionally with GIS and IoT teams to evaluate immersive technologies for future city planning.
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PROJECTS

Job Scraper Pipeline – Multi-Cloud AI Job Intelligence | *Python, GCP Cloud Functions, Cloud Scheduler, Groq Llama 3.3 70B, AWS Lambda, S3, SES, DynamoDB, Azure Cosmos DB, React, Vercel, GitHub Actions*

- Engineered a fully automated multi-cloud pipeline ingesting 30–50 job listings/day from USAJobs, Remotive, and Jobicy via GCP Cloud Functions on a daily Cloud Scheduler trigger.
- Integrated Groq Llama 3.3 70B for AI enrichment and relevance scoring; Azure Cosmos DB handles cross-run deduplication; enriched JSON stored to timestamped S3 keys; DynamoDB provides two-step telemetry.
- AWS Lambda delivers a daily HTML email digest via SES with heartbeat alerts on empty-run days; Pydantic validates all Groq output; full CI/CD via GitHub Actions deploying to AWS and GCP.
- React/Vercel dashboard at job-scraper-dashboard-taupe.vercel.app · Repo: github.com/nakucoder/job-scraper-pipeline

Athletic Intelligence – AI Fitness Coaching Mobile App | *React Native, Expo, Node.js, Express, Prisma, PostgreSQL (Supabase), Groq Llama 3.3 70B, Docker, Render, GitHub Actions*

- Built a full-stack mobile fitness app with dual AI personas — OLI (performance mode) and EMI (recovery mode) — powered by Groq’s Llama 3.3 70B for personalized coaching conversations.
- Implemented a deterministic readiness scoring engine computing fatigue and monotony metrics from logged workout history to dynamically guide AI recommendations.
- Designed a Weekly Planning System with WeeklyPlan/DailyAssignment data models, five REST endpoints, calendar UI with dot indicators, and chat-to-calendar conflict detection.
- Backend auto-deployed on Render via Docker with Prisma ORM over Supabase PostgreSQL; live device testing via Expo Go on Android.

Miami Data Dashboard + Pipeline – Serverless Real-Time Data Platform | *Python, FastAPI, AWS Lambda, EventBridge, API Gateway, S3, CloudWatch, SNS, React, Vite, Recharts, Vercel*

- Migrated three data pipelines (weather, crypto, stock) from EC2/Docker to fully serverless AWS Lambda triggered by EventBridge, eliminating idle compute costs.
- S3 cache pattern with materialized history.json dramatically reduced API response times; six API Gateway endpoints serve the React/Recharts dashboard at miami-dashboard-plum.vercel.app.
- Hardened with CloudWatch alarms, SNS alerts, Lambda timeout enforcement, and maximum-retry-attempts 0 to prevent runaway costs and ensure full observability.

AI Gateway Engine | *Node.js, TypeScript, JWT, Rate Limiting, Python, REST API*

- Built a high-performance API Gateway in Node.js/TypeScript bridging a React frontend and Python AI Engine, featuring JWT authentication, rate limiting, and real-time health monitoring.
- Designed to handle concurrent requests with structured error handling and endpoint-level access control; deployed with CI/CD via GitHub Actions. Repo: github.com/nakucoder/ai-gateway-engine

Tetris Clone | *Python, Pygame*

- Developed a classic Tetris game from scratch implementing collision detection, grid-based state management, line-clearing algorithms, and real-time input handling. Repo: github.com/nakucoder/Python-Tetris

EDUCATION

Miami Dade College

Bachelor of Science in Software Engineering

Relevant Coursework: Data Structures, Algorithms, Software Engineering I & II, Database Design

December 2025